

BEC503 Digital Communication — syllabus extract

Extracted verbatim from vtu-3ecesy11-2022.pdf (313,731 characters of machine-readable text).

-OEC (4 Credits) template

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DIGITAL COMMUNICATION

Semester

5

Course Code

BEC503

CIE Marks

50

Teaching Hours/Week (L:T:P: S)

4:0:0:0

SEE Marks

50

Total Hours of Pedagogy

50 Hours

Total Marks

100

Credits

04

Exam Hours

3 Hours

Examination type (SEE)

Theory

Course objectives:

x

Understand the concept of signal processing of digital data and signal conversion to symbols at the transmitter and receiver.

x

Compute performance metrics and parameters for symbol processing and recovery in ideal and corrupted channel conditions. x

Understand the principles of spread spectrum communications.

x

Understand the basic principles of information theory and various source coding techniques.

x

Build a comprehensive knowledge about various Source and Channel Coding techniques.

x

Discuss the different types of errors and error detection and controlling codes used in the communication channel.

x

Understand the concepts of convolution codes and analyze the code words using time domain and transform domain approach.

Teaching-Learning Process (General Instructions) These are sample Strategies, which teachers can use to accelerate the attainment of the various course outcomes.

1.

Lecture method (L) does not mean only the traditional lecture method, but a different type of teaching method may be adopted to develop the outcomes.

2.

Arrange visits to nearby PSUs such as BHEL, BEL, ISRO, etc., and small-scale communication industries.

3.

Show Video/animation films to explain the functioning of various modulation techniques, Channel, and source coding.

4.

Encourage collaborative (Group) Learning in the class

5.

Ask at least three HOTS (Higher-order Thinking) questions in the class, which promotes critical thinking

6.

Adopt Problem Based Learning (PBL), which fosters students

9

Analytical skills, develop thinking skills such as the ability to evaluate, generalize & analyze information rather than simply recall it.

7.

Topics will be introduced in multiple representations.

8.

Show the different ways to solve the same problem and encourage the students to come up with their own creative ways to solve them.

9.

Discuss how every concept can be applied to the real world - and when that's p

ossible, it helps improve the students' understanding.

Module-1

Bandpass Signals to Equivalent Lowpass:

Hilbert Transform, Pre-envelopes, Complex envelopes of Band-pass Signals, Canonical Representation of Bandpass signals. Signalling over AWGN Channels- Introduction, Geometric representation of signals, Gram- Schmidt Orthogonalization procedure, Conversion of the continuous AWGN channel into a vector channel , Optimum receivers using coherent detection: ML Decoding, Correlation receiver, matched filter receiver.

Module-2

Digital Modulation Techniques:

Phase shift Keying techniques using coherent detection: generation, detection

and error probabilities of BPSK and QPSK, M

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ary PSK, M

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ary QAM. Frequency shift keying techniques using

Coherent detection: BFSK generation, detection and error probability. BFSK using Noncoherent Detection, Differential Phase Shift Keying.

Module-3

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Information theory:

Introduction, Entropy, Source Coding Theorem, Lossless Data Compression

Algorithms, Discrete Memoryless Channels, Mutual Information, Channel capacity, Channel Coding

Theorem, Information Capacity Law (Statement).

Module-4

Error Control Coding:

Error Control Using Forward error Correction,

Linear Block Codes:

Definitions, Matrix Descriptions, Syndrome and its properties, Minimum distance

Considerations, Syndrome Decoding, Hamming Codes. Cyclic Codes:

Properties, Generator and Parity Check Polynomial and matrices, Encoding, Syndrome

computation, Examples.

Module-5

Convolutional Codes:

Convolutional Encoder, Code tree, Trellis Graph and State graph, Recursive systematic

Convolutional codes, Optimum decoding of Convolutional codes, Maximum Likelihood Decoding of Convolutional codes: The Viterbi Algorithm, Examples.

Course outcome (Course Skill Set) At the end of the course, the student will be able to :
1.

Apply the concept of signal conversion to vectors in communication transmission and reception.
2.

Perform the mathematical analysis of digital communication systems for different modulation techniques.
3.

Apply the Source coding and Channel coding principles for the discrete memoryless channels.
4.

Compute the codewords for the error correction and detection of a digital data using Linear Block Code, Cyclic Codes and Convolution Codes.
5.

Design encoding and decoding circuits for Linear Block Code, Cyclic Codes and Convolution Codes.

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Assessment Details (both CIE and SEE) The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks

out of 50)

and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks).

The

student is declared as a pass in the course if he/she secures a minimum of 40%

(40 marks out of

100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together. Continuous Internal Evaluation: x

There are 25 marks for the CIE's Assignment component and 25 for the Internal Assessment Test component.

x

Each test shall be conducted for 25 marks. The first test will be administered after 40-50% of the coverage of the syllabus, and the second test will be administered after 85-90% of the coverage of the syllabus. The average of the two tests shall be scaled down to 25 marks

x

Any two assignment methods mentioned in the 220B4.2, if an assignment is project-based then

only one assignment for the course shall be planned. The schedule for assignments shall be planned properly by the course teacher. The teacher should not conduct two assignments at the end of the semester if two assignments are planned. Each assignment shall be conducted for 25 marks. (If two assignments are conducted then the sum of the two assignments shall be scaled down to 25 marks)

x

The final CIE marks of the course out of 50 will be the sum of the scale-down marks of tests and assignment/s marks.

Internal Assessment Test question paper is designed to attain the different levels of Bloom

ii

s taxonomy

as per the outcome defined for the course. Semester-End Examination: Theory SEE will be conducted by University as per the scheduled timetable, with common question papers for the course (duration 03 hours).

1.

The question paper will have ten questions. Each question is set for 20 marks.

2.

There will be 2 questions from each module. Each of the two questions under a module (with a maximum of 3 sub-questions), should have a mix of topics under that module.

3.

The students have to answer 5 full questions, selecting one full question from each module

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4.

Marks scored shall be proportionally reduced to 50 marks.
Suggested Learning Resources: Text Book

1.

Simon Haykin,

3

Digital Communication Systems

,

, John Wiley & sons, 2014, ISBN 978-81

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265

-4231-

4.

Reference Books

1.

B.P Lathi, Zhi Ding,

20

Modern Digital and Analog Communication Systems

26

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th

Edition, Oxford

University press, ISBN: 9780198073802, 2017

2.

K Sam Shanmugam,

3

Digital and analog communication systems

,

, Wiley India Pvt. Ltd,

2017,

ISBN:978-

81

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265

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3680

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1,

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3.

K.N Hari Bhat, D. Ganesh Rao,

3

Information Theory and Coding

,

, Cengage Learning India Pvt Ltd,

2017

,

ISBN: 93-

866

-5092-

4,

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Web links and Video Lectures (e-Resources):

1.

Principles of Communication Systems Part II

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https://onlinecourses.nptel.ac.in/noc19_ee47/preview

Activity Based Learning (Suggested Activities in Class)/ Practical Based learnin

g

1.

Experiential Learning by using free and open source software

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s SCILAB or OCTAVE or Python

Annexure-

II

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